Year 9 Mathematics

Unit 2: Area and Volume

2.5 Cylinders

Cylinders

STARTER · 5 MIN

Date: _____ **Title: Cylinders**

LEARNING OBJECTIVES

- Calculate the volume of a cylinder
- Calculate the surface area of a cylinder
- Solve problems with cylinders

STARTER *Answer in your book. No calculators.*

1. Area of a circle, radius 3 cm (1 d.p.)
2. Circumference, radius 3 cm (1 d.p.)
3. How many litres is 1000 cm^3?
4. Volume of a prism: area 12 cm^2, length 5 cm

Starter — Answers

STARTER

Tick or correct in green pen

1. **28.3 cm²**
2. **18.8 cm**
3. **1 litre**
4. **60 cm³**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can calculate the volume of a cylinder	☐
SILVER	I can calculate the surface area of a cylinder	☐
GOLD	I can solve problems with cylinders	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Cylinders

INTRODUCTION · 3 MIN

A cylinder is a prism with a circle as its cross-section.

Volume

Area of the circle × height.

$V = \pi r^2 h$

Curved surface

Unrolls into a rectangle, C long.

curved area $= 2\pi rh$

Total surface area

Two circles plus the curved surface.

$SA = 2\pi r^2 + 2\pi rh$

Capacity

Change cm^3 into litres.

$1000\ cm^3 = 1$ litre

Examples 1 and 2 — Volume of a Cylinder

EXAMPLES · 5 MIN

BRONZE

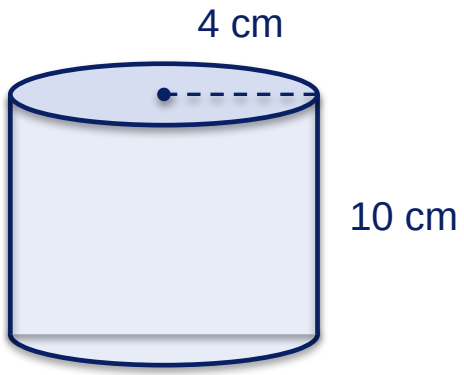
Watch and copy into your book

Hint $V = \pi r^2 h$ $r = d \div 2$

Example 1

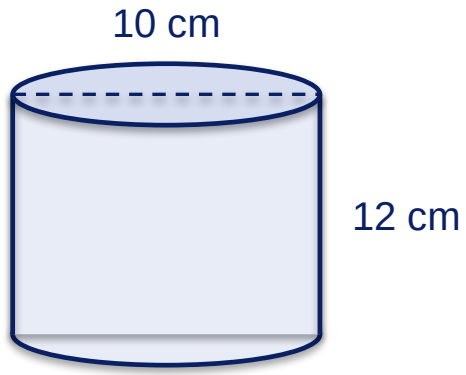
Find the volume.

Give your answer to 1 d.p.



Example 2

Find the volume to the nearest whole number.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$V = \pi r^2 h$

$V = \pi \times 4^2 \times 10$

$= 502.65\ldots$

$= 502.7\ \text{cm}^3$ (1 d.p.)

Tip: r^2 first, then $\times \pi \times h$.

Example 2

Diameter 10, so r = 5

$V = \pi \times 5^2 \times 12$

$= 942.47\ldots$

$= 942\ \text{cm}^3$

Tip: nearest whole number: no decimals.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the volume. Give your answers to 1 d.p.

Hint $V = \pi r^2 h$ $r = d \div 2$

1. radius 3 cm, height 8 cm
2. radius 5 cm, height 4 cm
3. diameter 6 cm, height 10 cm
4. diameter 12 cm, height 5 cm
5. radius 2.5 cm, height 20 cm

★ radius 14 m, height 7.5 m (nearest m^3)

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $226.2\ cm^3$
2. $314.2\ cm^3$
3. $282.7\ cm^3$
4. $565.5\ cm^3$
5. $392.7\ cm^3$

★ $4618\ m^3$

Examples 3 and 4 — Surface Area of a Cylinder

EXAMPLES · 5 MIN

SILVER

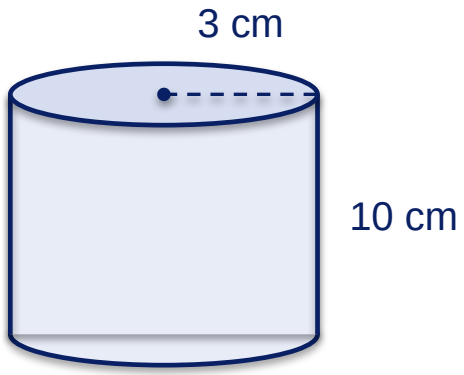
Watch and copy into your book

Hint SA = $2\pi r^2 + 2\pi rh$ curved surface = $2\pi rh$

Example 3

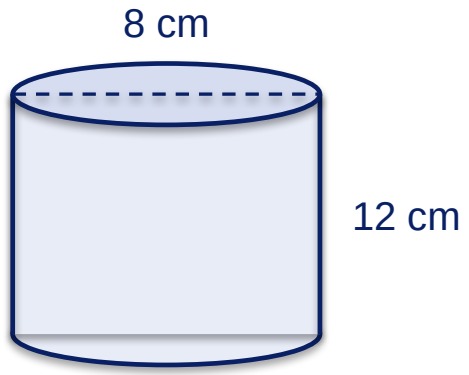
Find the total surface area.

Give your answer to 1 d.p.



Example 4

A label covers the curved surface of a can. Find its area to 1 d.p.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

2 circles + curved surface

$2 \times \pi \times 3^2 = 56.54\ldots$

$2 \times \pi \times 3 \times 10 = 188.49\ldots$

SA = 245.0 cm² (1 d.p.)

Tip: add before rounding.

Example 4

Curved surface only: πdh

$\pi \times 8 \times 12$

$= 301.59\ldots$

= 301.6 cm² (1 d.p.)

Tip: the label has no top or bottom.

Task 2

PRACTICE · 9 MIN

SILVER

Questions 1–3, 6: total surface area. 4–5: curved surface only. 1 d.p.

Hint
SA = $2\pi r^2 + 2\pi rh$
curved surface = $2\pi rh$

1 radius 2 cm, height 5 cm

2 radius 5 cm, height 10 cm

3 diameter 10 cm, height 4 cm

4 Curved: radius 4 cm, height 15 cm

5 Curved: diameter 7 cm, height 10 cm

★ radius 3.5 cm, height 12 cm

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 88.0 cm²
2. 471.2 cm²
3. 282.7 cm²
4. 377.0 cm²
5. 219.9 cm²
★ 340.9 cm²

Examples 5 and 6 — Cylinder Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $V = \pi r^2 h$ $h = V \div \pi r^2$ $1000\ cm^3 = 1$ litre

Example 5

(3 marks)

A cylindrical tank has radius 20 cm and height 50 cm.

How many litres does it hold?

Give your answer to 1 d.p.

Example 6

(3 marks)

A cylinder has volume $500\ cm^3$ and radius 5 cm.

Find its height to 1 d.p.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Volume, then ÷ 1000

$\pi \times 20^2 \times 50 = 62\,831.8\ldots\ cm^3$

$62\,831.8\ldots \div 1000$

= 62.8 litres

Tip: 1000 cm^3 = 1 litre

Example 6

$h = V \div \pi r^2$

$\pi \times 5^2 = 78.539\ldots$

$500 \div 78.539\ldots$

= 6.4 cm (1 d.p.)

Tip: check: $\pi \times 25 \times 6.4 \approx 500$

Task 3

PRACTICE · 9 MIN

GOLD

Give your answers to 1 d.p. unless told otherwise

Hint $V = \pi r^2 h$ $h = V \div \pi r^2$
$1000\ cm^3 = 1$ litre

1. r = 10 cm, h = 30 cm. Capacity in litres?
2. d = 30 cm, h = 40 cm. Capacity in litres?
3. $V = 1000\ cm^3$, r = 6 cm. Find h.
4. $V = 750\ cm^3$, r = 4 cm. Find h.
5. V = 2 litres, r = 7 cm. Find h.

★ r = 15 cm, h = 40 cm. How many full 1-litre bottles can it fill?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **9.4 litres**
2. **28.3 litres**
3. **8.8 cm**
4. **14.9 cm**
5. **13.0 cm**
- ★ **28 274 cm^3: 28 bottles**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Volume: radius 2 cm, height 10 cm (1 d.p.)

SILVER Curved surface: radius 3 cm, height 5 cm (1 d.p.)

GOLD $V = 400\ cm^3$, $r = 4$ cm. Find h (1 d.p.).

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **125.7 cm^3**

SILVER **94.2 cm^2**

GOLD **8.0 cm**

Exam Practice — Volume of a Cylinder

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

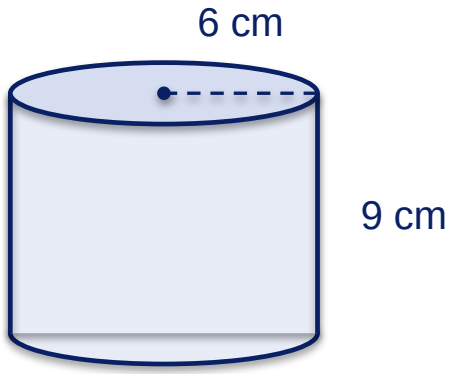
Hint $V = \pi r^2 h$ $r = d \div 2$

Question 1

(2 marks)

Find the volume of this cylinder.
Give your answer to 1 d.p.

Diagram NOT accurately drawn



..... cm^3

Question 2

(3 marks)

A cylinder has radius 12 m and height 4.5 m. Calculate its volume to the nearest whole number.

..... m^3

Exam Practice — Surface Area

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

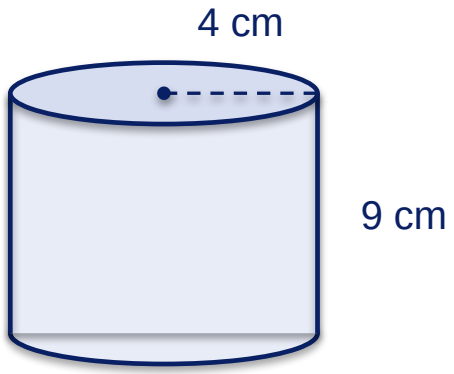
Hint SA = $2\pi r^2 + 2\pi rh$ curved surface = $2\pi rh$

Question 3

(3 marks)

Find the total surface area.
Give your answer to 1 d.p.

Diagram NOT accurately drawn



..... cm^2

Question 4

(2 marks)

A can has diameter 7.5 cm and height 11 cm. A label covers the curved surface. Find its area to 1 d.p.

..... cm^2

Exam Practice — Cylinder Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $V = \pi r^2 h$ $h = V \div \pi r^2$ $1000\ cm^3 = 1$ litre

Question 5

(3 marks)

A cylindrical water tank has radius 25 cm and height 80 cm.
How many litres does it hold?
Give your answer to 1 d.p.

..... litres

Question 6

(3 marks)

A drinks can is a cylinder with volume $330\ cm^3$ and radius 3.2 cm.
Find its height to 1 d.p.

..... cm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$\pi \times 6^2 \times 9$ — M1

$= 1017.9\ cm^3$ — A1

Tip: $V = \pi r^2 h$

Question 2

$\pi \times 12^2 = 452.38\ldots$ — M1

$\times 4.5 = 2035.75\ldots$ — M1

$= 2036\ m^3$ — A1

Tip: round only at the end.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$2 \times \pi \times 4^2 = 100.53\ldots$ **M1**

$2 \times \pi \times 4 \times 9 = 226.19\ldots$ **M1**

$= 326.7\ \text{cm}^2$ **A1**

Tip: two circles and the curved surface

Question 4

$\pi \times 7.5 \times 11$ **M1**

$= 259.2\ \text{cm}^2$ **A1**

Tip: curved surface = πdh

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$\pi \times 25^2 \times 80$	M1
$= 157\,079.6\ldots\ cm^3$	A1
= 157.1 litres	A1

Tip: ÷ 1000 to change cm^3 to litres

Question 6

$\pi \times 3.2^2 = 32.16\ldots$	M1
$330 \div 32.16\ldots$	M1
= 10.3 cm	A1

Tip: $h = V \div \pi r^2$